Phase Detector

FIR:

Timing:

Clock	Target	Estimated	Uncertainty
ap_clk	10.00 ns	7.256	2.70 ns

Latency:

Latency (cycles)		Latency (absolute)		Interval		Pipeline Type
min	max	min	max	min	max	
190	190	1.900 us	1.900 us	2	2	yes

1. Phase Detector Throughput

$$\frac{1}{2 \times 7.256 \times 10^{-9}} = 68.9MHz$$

We previously used *complete* partitioning, which utilized > 100% LUTs. Switching it to cyclic partitioning with a factor of 16 brought that down. The initiation interval for our implementation *cordiccart2pol* is capped at 2 for our current implementation, so that is our bottleneck.

FIR is fully pipelined, so with more memory access, we could get that to II=1. The biggest hurdle would be getting CORDIC to II=1. It currently has many logic steps (one initial, and one per iteration) which are costly. Reducing it to II=1 would likely require fewer iterations.

A different approach would be to make a duplicate hardware path and alternate between the two of them. A quick google search indicates that this is called a *ping-pong buffer*. It would require twice the CORDIC hardware and some MUXs.

Github Repository

https://github.com/ShantonuM/wes237c_fa2026/tree/main/proj2

https://github.com/UCSD-237C-2026/wes-237c-project-2-cordic-group-10

Video Link

https://youtu.be/GH48yIaaCwk